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Manual

LXL Series:
AK-Dinrail-LXL
ATTO-LXL



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Setup

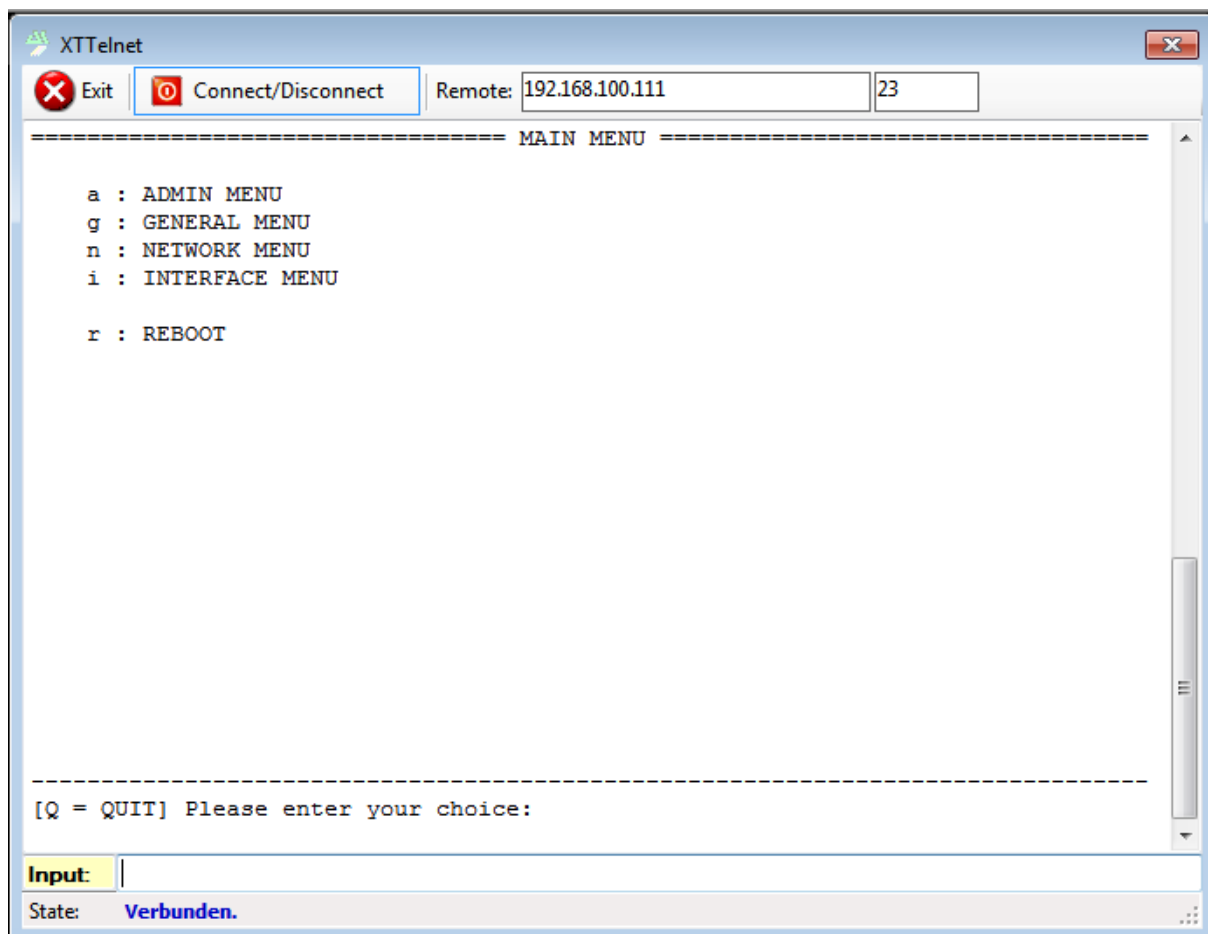
Access via Network

The devices from AK-Nord are delivered from the factory with the DHCP setting activated. I.e. you need a started DHCP server in your network, so that a LXL device gets a valid IP from this server.

To determine which IP address the device has received, please install the XTAdminXXL tool from AK-Nord. You can find this on our website www.ak-nord.de in the download area.

If you do not want to install the XTAdminXXL tool, you can alternatively look up the assigned IP address in the logs of the DHCP server using the MAC address of the device.

If you do not want to install the XTAdminXXL tool, you can alternatively look up the assigned IP address in the logs of the DHCP server using the MAC address of the device.





Once you know the assigned IP address of the device, you can configure it via Telnet menu (port 23).

For this purpose please use the Telnet window in our XTAdminXXL tool or one of the other Telnet programs like e.g. PuTTY.

For devices without root access, you can log in with the user admin. The default password for this is ak-nord. For root image, the default password for user root is xt. As soon as you are logged in to the device via ssh, you can access the configuration menu using the telnet localhost command.

Access via debug console (UART):

Devices such as SER2USB-LXL (without WLAN) do not have a network interface, but you can use a USB-C cable to connect the device to be configured to a PC via USB and use a program such as putty to establish a connection via the serial interface. As soon as the SER2USB device is connected to the Windows PC, a new COM port appears in the Windows device manager. You will need the COM number assigned by Windows, e.g. COM10, to establish the connection. Then set the following connection parameters in putty (or another program for serial communication):

COM no. e.g. COM10 (this can be found in the Windows device manager)

Baud rate: 115200

Data bits: 8

Parity: N

Stop bits: 1

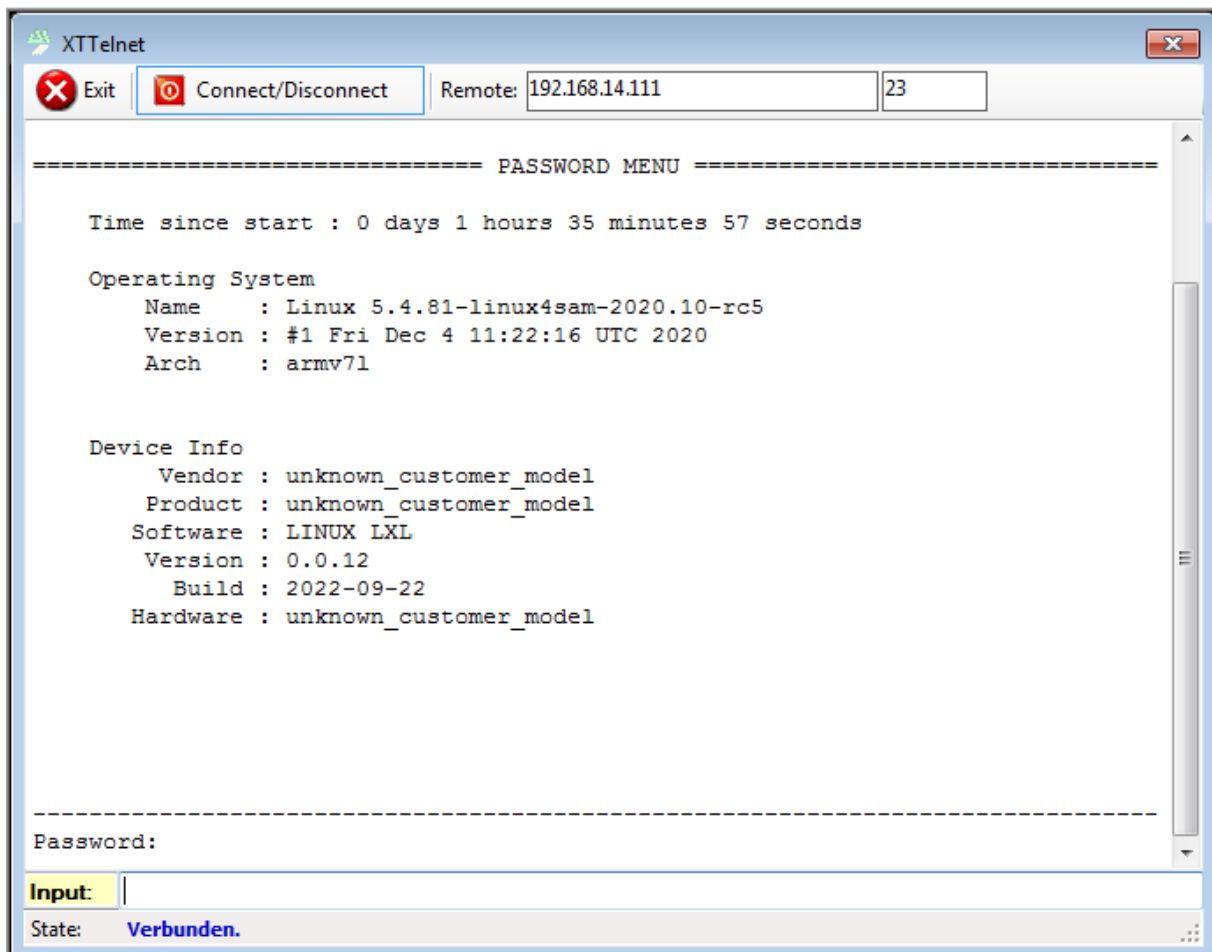
Flow control: N

As soon as the connection is established, you will see the login prompt and can log in with the same users as for ssh access, see previous chapter. After successfully logging in, you can access the configuration menu via the telnet localhost command.



Telnet-Configuration

The first thing you will see is the password window. The default password is **xt**. For security reasons, you should change this password during the initial configuration.



After entering the password correctly, you will get to the main menu. Navigation through the Telnet menu is very simple. You have a menu item ID in front of each menu item. This is usually a letter or a number like "a" for the "ADMIN MENU" or "1" for the hostname:

a : ADMIN MENU"
1 : Hostname = USB-SERVER-LXL

Enter "a" to call up the ADMIN MENU. Press "q" to close the current menu and return to the previous menu. Entering "q" in the MAIN MENU terminates the Telnet connection to the device. To change a configuration value, you must enter the <menu item ID>=<new value>. For the host name, for example, it would look like this:

1=usbserver_webcam



12.11.2022

MAIN MENU:

The screenshot shows the XTtelnnet application window. The title bar reads 'XTtelnnet'. Below the title bar is a toolbar with three buttons: 'Exit' (red X icon), 'Connect/Disconnect' (red circle with white 'O' icon), and a 'Remote:' field containing '192.168.14.111' and a port field containing '23'. The main text area displays the following menu:

```
===== MAIN MENU =====  
  
a : ADMIN MENU  
g : GENERAL MENU  
n : NETWORK MENU  
i : INTERFACE MENU  
h : HARDWARE MENU  
  
-----  
[Q = QUIT] Please enter your choice:  
  
Input: |  
State:  Verbunden.
```

The 'Input:' field is currently empty, and the 'State:' is 'Verbunden.' (Connected).

**ADMIN MENU:**

XTtelnnet

Exit Connect/Disconnect Remote: 192.168.100.111 23

===== ADMIN MENU =====

1 : Telnet port = 23

2 : Telnet password = **

3 : Reset Server(Y/N) = N

4 : Resetserver password =

a : Apps

b : Apps autostart

c : Services

d : Services autostart

[Q = QUIT] Please enter your choice:

Input:

State: **Verbunden.**

1 : Telnet port

Here you can configure the port for network access. Default is 23.

2 : Telnet password

Here you can change the password for Telnet access.

3 : Reset server

You can enable a reset server here. This allows you to reboot the entire device via TCP port 8888 or restart individual bridge services via the configured RESET_PORT in the respective bridge.

4 : Resetserver password

If a password has been configured here, it must be entered after successfully connecting to the reset ports (8888 or the respective bridge reset port).

a : Apps

Here you can start or stop the AK-Nord programs.



12.11.2022

b : Apps autostart

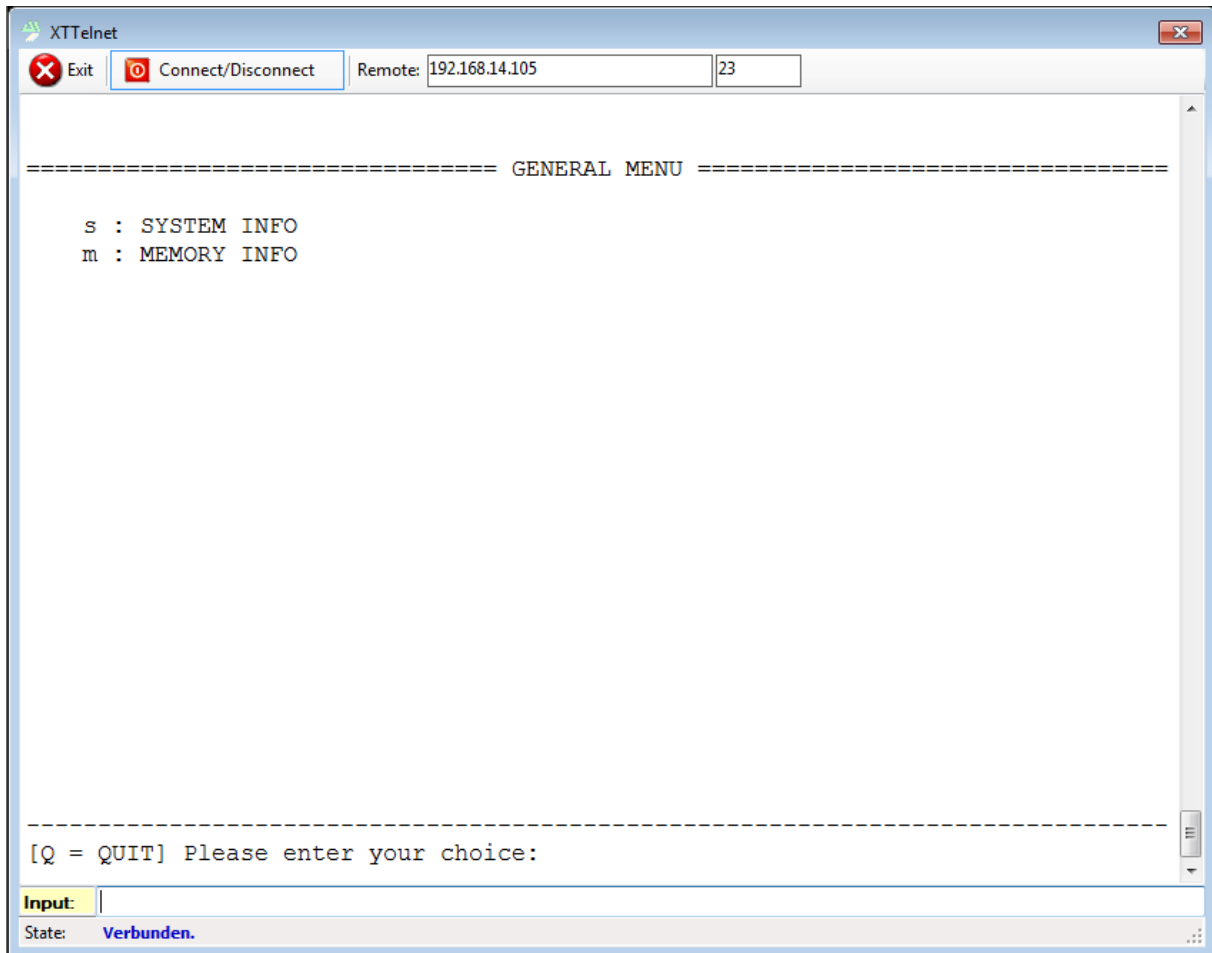
Configuration of Autostart when booting the AK-Nord programs.

c : Services

Here you can start or stop Linux services like e.g. NTP server.

d : Services autostart

Here you can configure the autostart of Linux services at boot time.

**GENERAL MENU:****s : SYSTEM INFO**

Here you can view the system information like Linux version, runtime etc..

m : MEMORY INFO

Here information about the consumption of the working memory is displayed.



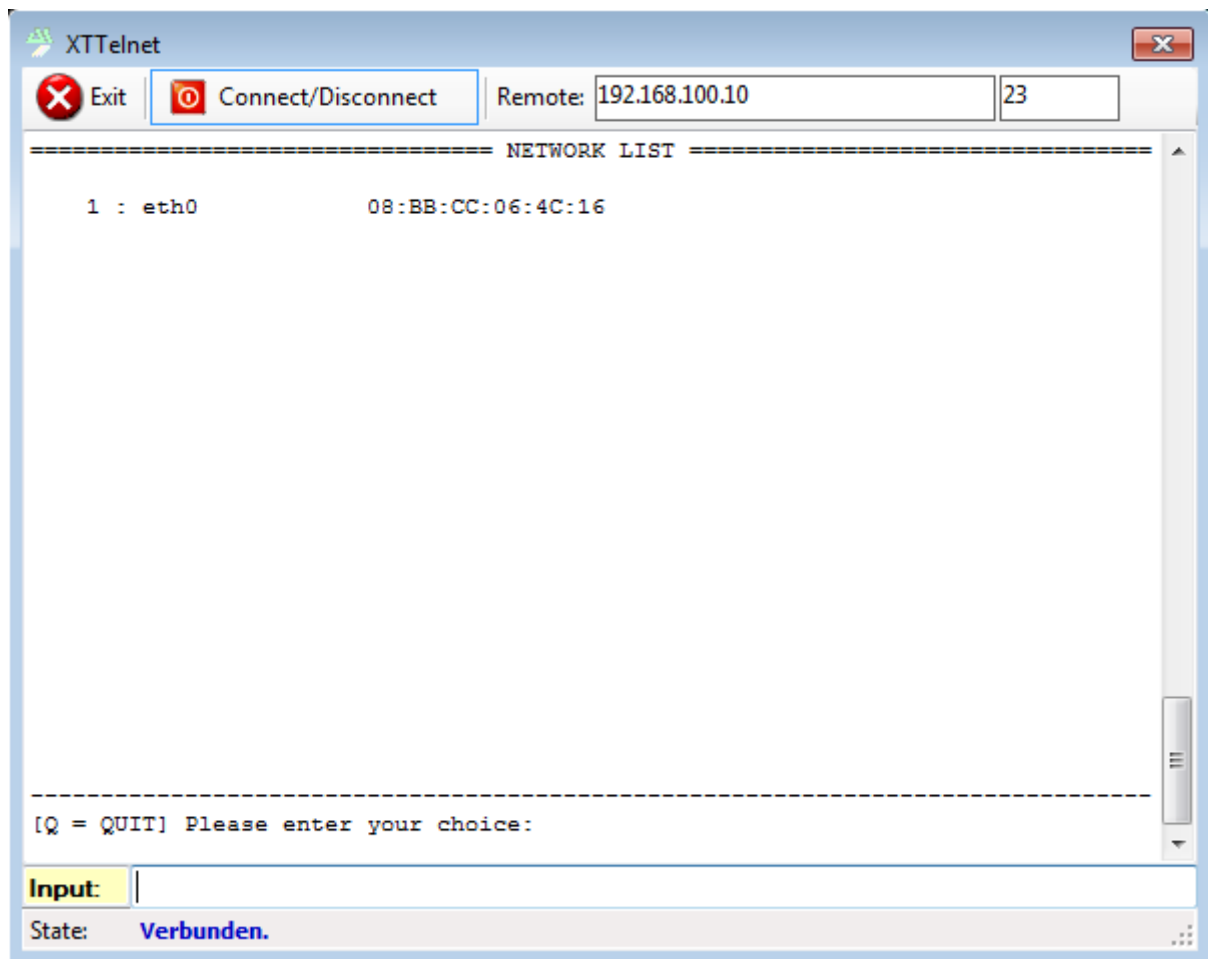
12.11.2022

NETWORK MENU:

The screenshot shows the XTtelnnet application window. The title bar reads 'XTtelnnet'. Below the title bar, there is a toolbar with three buttons: 'Exit' (with a red 'X' icon), 'Connect/Disconnect' (with a red 'O' icon), and a 'Remote:' field containing '192.168.100.10' and a port field containing '23'. The main area of the window displays a menu titled 'NETWORK MENU' with the following options:

- 1 : NETWORK LIST
- 2 : DNS MENU
- 3 : NTP MENU
- 4 : TCP Info
- 5 : UDP Info

Below the menu, a dashed line separates it from the prompt: '[Q = QUIT] Please enter your choice:'. At the bottom, there is an 'Input:' field and a 'State:' field showing 'Verbunden.'.

**NETWORK LIST:**

Here you can see the list of all available network cards.



NETWORK CARD CONFIGURATION:

The screenshot shows the XTTelnet application window. At the top, there is a title bar with 'XTTelnets' and a close button. Below the title bar, there is a toolbar with 'Exit' (red X icon), 'Connect/Disconnect' (red O icon), and a 'Remote:' field containing '192.168.100.10' and a port field containing '23'. The main area displays the configuration for 'eth0' with the following settings:

```
----- eth0 -----  
  
d : DHCP (Y/N)    = Y  
  
1 : IPv4 address  = 192.168.100.10  
2 : Subnet mask   = 255.255.0.0  
3 : Gateway       =  
  
Nameserver1      = 127.0.0.1  
Nameserver2      =  
  
MAC address      = 08:BB:CC:06:4C:16  
  
-----  
[Q = QUIT] Please enter your choice:  
  
Input: |  
State:  Verbunden.
```

d : DHCP(Y/N)

Here you can configure whether the IP address should be obtained dynamically from a DHCP server or not.

1 : IPv4 address

If DHCP is active, the current IPv4 address of the device is displayed here. If DHCP is switched off, the fixed IP address for the device can be assigned here.

2 : Subnet mask

If DHCP is active, the network mask assigned by the DHCP server is displayed here. If DHCP is switched off, the network mask can be assigned manually here..

3 : Gateway

A gateway can be configured manually here if DHCP is off. When DHCP is enabled, the gateway assigned by the DHCP server is displayed here.

**DNS MENU:**

The screenshot shows the XTTelnet application window. At the top, there is a title bar with the text 'XTTelnets'. Below the title bar, there is a toolbar with three buttons: 'Exit' (with a red X icon), 'Connect/Disconnect' (with a red circle and a white 'C' icon), and a 'Remote:' field containing the IP address '192.168.14.105' and a port field containing '23'. The main area of the window displays the 'DNS MENU' with the following options:

```
===== DNS MENU =====  
  
1 : Hostname      = USB-SERVER-LXL  
  
2 : Nameserver1 =  
3 : Nameserver2 =  
  
-----  
[Q = QUIT] Please enter your choice:  
  
Input:   
State:  Verbunden.
```

1 : Hostname

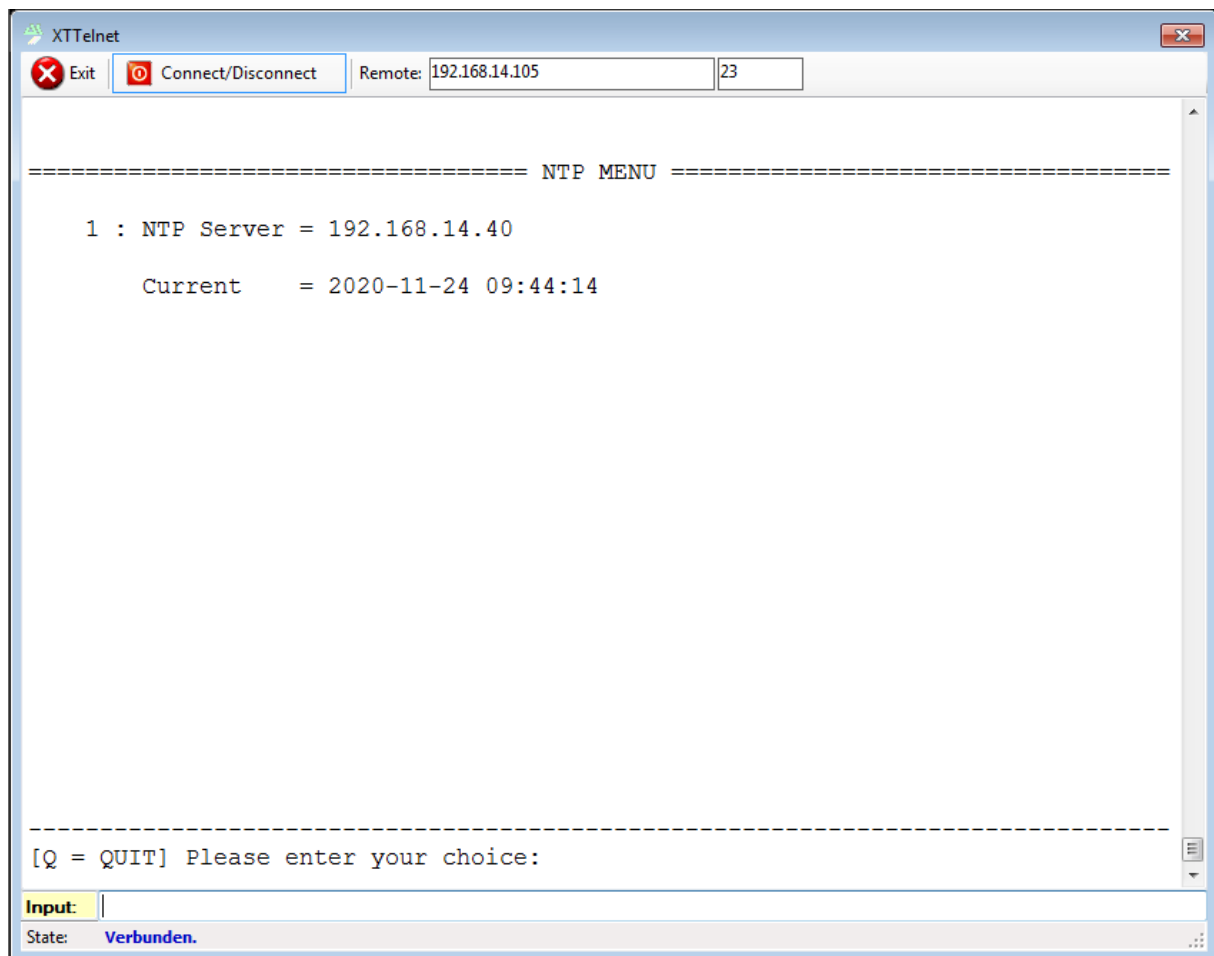
Here you can configure the host name of the device. The following characters are allowed: 0-9, A-z, minus sign and dot.

2 : Nameserver1

Here you can enter IP address of your 1st name server manually.

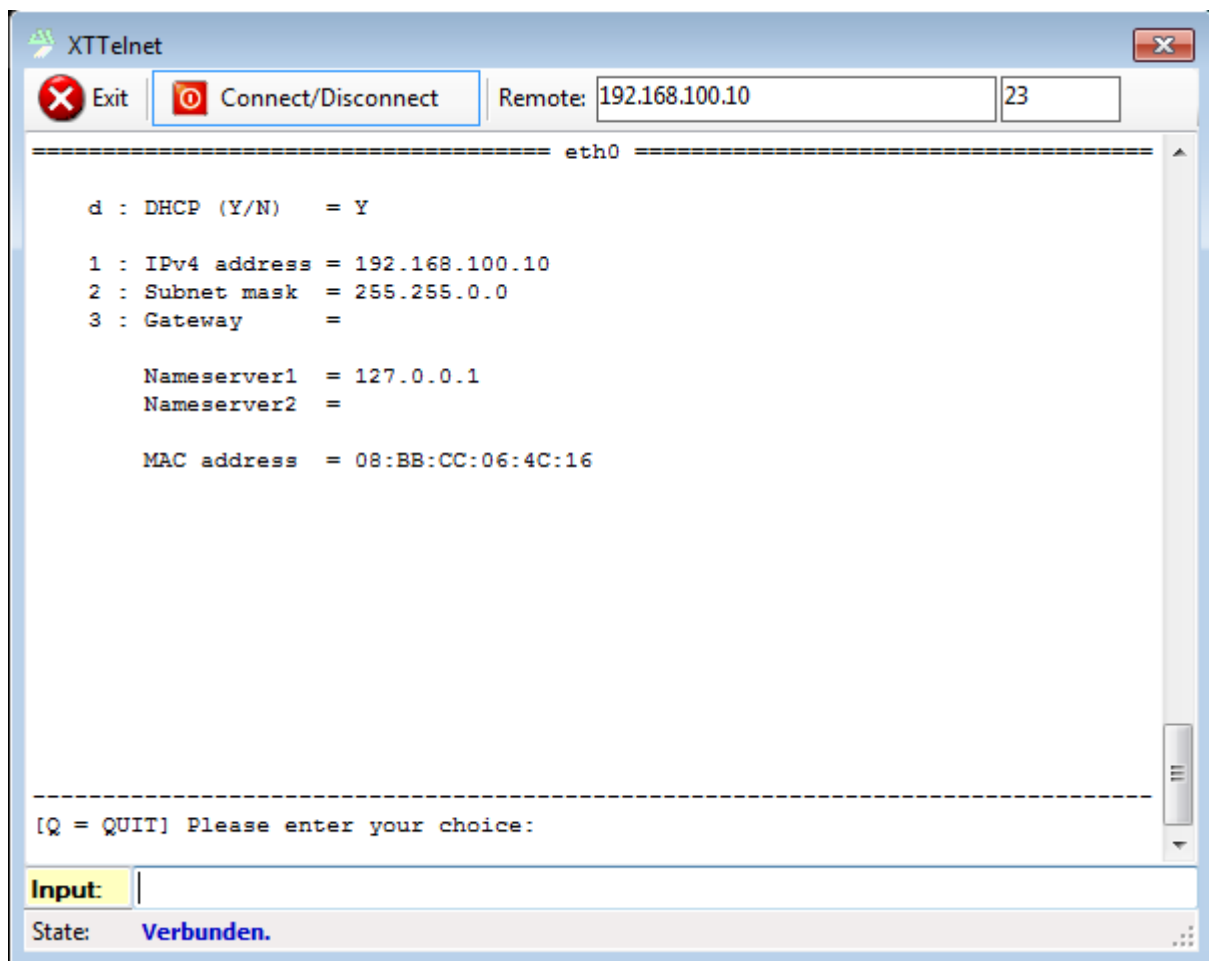
3 : Nameserver2

Here you can enter IP address of your 2nd name server manually.

**NTP MENU:****1 : NTP Server**

Here you can enter the IP address or the hostname for the NTP server for time synchronization.

Current displays the current time on the device.

**TCP INFO:**

Here you can see an overview of all active network connections of the device via TCP.

**UDP INFO:**

The screenshot shows the XTTelnet application window. At the top, there is a title bar with the text 'XTTelnets'. Below the title bar, there is a menu bar with 'Exit' and 'Connect/Disconnect' buttons. To the right of the menu bar, there is a 'Remote:' label followed by a text box containing '192.168.100.10' and a port box containing '23'. The main area of the window displays 'UDP Info' in a monospaced font. Below this, there is a table with the following columns: 'Local-IP', 'Port', 'Remote-IP', 'Port', and 'State'. The table contains three rows of data. Below the table, there is a status bar that says 'State: Verbunden.' and an 'Input:' field.

Local-IP	Port	Remote-IP	Port	State
192.168.100.010	123	000.000.000.000	0	7
192.168.100.010	137	000.000.000.000	0	7
192.168.100.010	138	000.000.000.000	0	7

Page: 1 / 1

[Q = QUIT] Please enter your choice:

Input:

State: **Verbunden.**

Here you can see an overview of all active network connections of the device via UDP.



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INTERFACE MENU:

The screenshot shows a Windows-style application window titled "XTtelnnet". At the top, there is a toolbar with an "Exit" button (red X icon) and a "Connect/Disconnect" button (red square with a white circle and a red dot). To the right of these buttons are two input fields: "Remote:" containing "192.168.100.10" and a port field containing "23". The main area of the window displays the "INTERFACE MENU" which lists three options: "1 : DEVICE CONFIGURATION", "2 : BRIDGE CONFIGURATION", and "3 : VIRTUALUSB CONFIGURATION". Below this menu, a dashed line separates it from a prompt that says "[Q = QUIT] Please enter your choice:". At the bottom of the window, there is an "Input:" field and a "State:" label showing "Verbunden." (Connected).

XTtelnnet

Exit Connect/Disconnect Remote: 192.168.100.10 23

===== INTERFACE MENU =====

1 : DEVICE CONFIGURATION
2 : BRIDGE CONFIGURATION
3 : VIRTUALUSB CONFIGURATION

[Q = QUIT] Please enter your choice:

Input: |

State: **Verbunden.**



BRIDGE concept:

AK-Nord's bridge concept offers you, the customer, the greatest possible flexibility. You can dynamically store up to 10 parallel bridge configurations. Each bridge configuration can link two DEVICE objects with each other and thus ensure bidirectional communication between the two objects. You can also link two devices of the same device type.

There are different device types that you can configure:

NET: network connection (TCPSERVER, TCPCLIENT, UDPSERVER, UDPCLIENT)

UART: RS232/RS485 serial device

USB: USB device of the following classes (CDC 0x02, printer 0x07, HID 0x03, CDC-Data 0x0a)

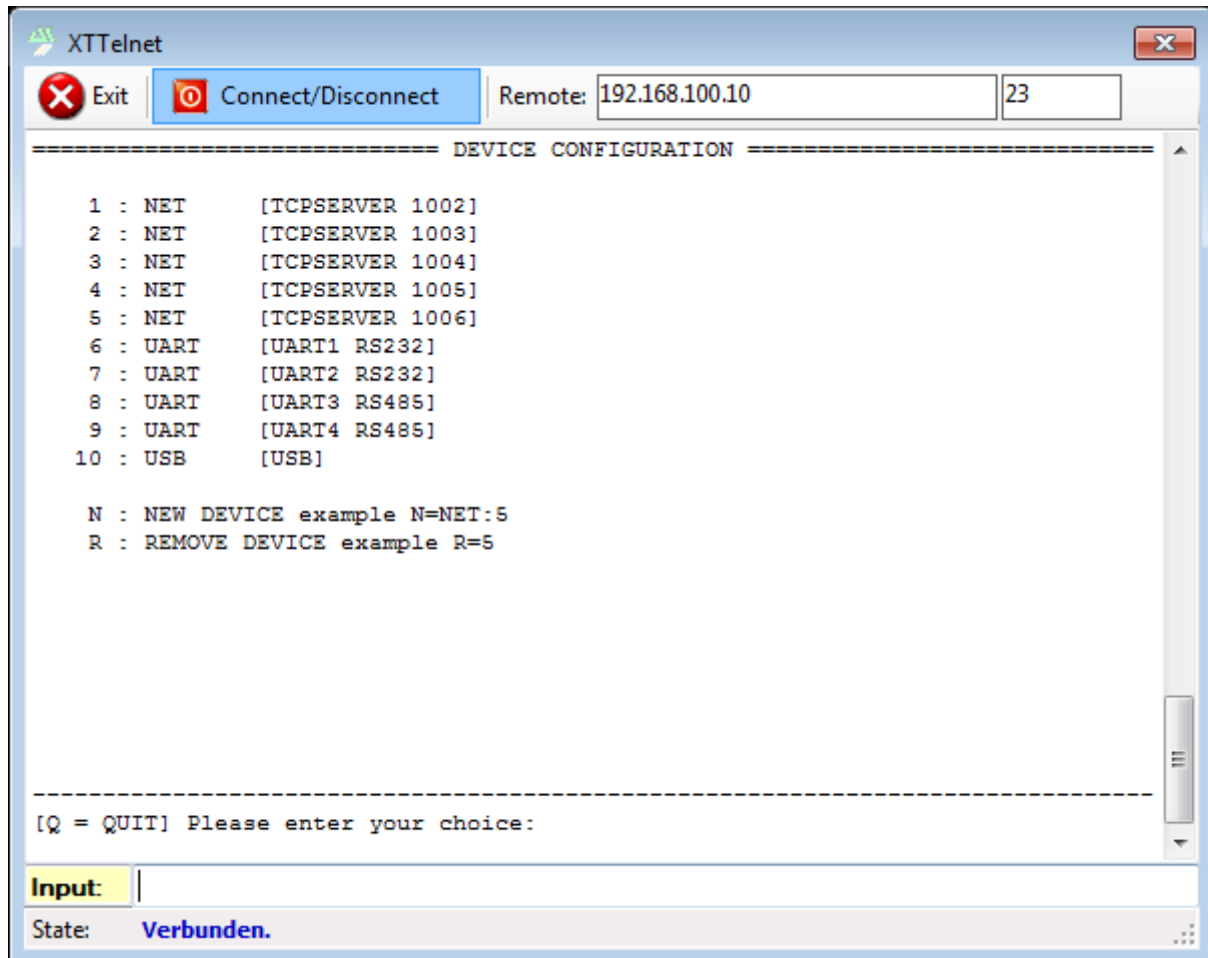
USBUART: USB2RS232 converter

FILE: File (only one direction: writing to the file)

TTLIO: Control of individual I/O pins

SPI: under development

I2C: under development

**DEVICE CONFIGURATION:****N : NEW DEVICE**

Here you can add a new device configuration. The type for the new device is mandatory so that suitable configuration parameters are added. If you enter N=UART, a new UART device is created with the next free DEVICE_ID.

Optionally, you can also specify a specific free ID. The input must then look like this: N=UART:11

R : REMOVE DEVICE

This allows you to delete a device configuration.

**UART DEVICE Parameter:**

The screenshot shows the XTTelnet application window. At the top, there is a title bar with the text 'XTTelnets'. Below the title bar, there is a menu bar with 'Exit' and 'Connect/Disconnect' buttons. To the right of the menu bar, there is a 'Remote:' label followed by two input fields: '192.168.100.10' and '23'. The main area of the window displays the configuration for 'DEVICE_6' as follows:

```
===== DEVICE_6 =====
1 : TYPE           = UART
2 : DESC           = UART1 RS232
3 : DEVPATH        = /dev/ttyS1
4 : BAUDRATE       = 9600
5 : DATABITS       = 8
6 : PARITY         = N
7 : STOPBITS       = 1
8 : FLOWCONTROL    = N
9 : INPUT_TIMEOUT  =
10 : RESET_TIMEOUT =
```

Below the configuration, there is a dashed line and the text '[Q = QUIT] Please enter your choice:'. At the bottom, there is an 'Input:' label followed by an empty text box. Below the text box, there is a 'State:' label followed by the text 'Verbunden.'.

1 : TYPE

Device type, the following values are permitted: NET, UART, USB, USBUART, TTLIO, SPI, I2C. Other types can be realised on request.
Note: If you change a configured type, all current settings are deleted, as each type requires its own configuration parameters.

2 : DESC

Description for this configuration.

3 : DEVPATH

UART device file under Linux, e.g. /dev/ttyS0 - /dev/ttyS5.

4 : BAUDRATE

Baudrate such as: 2400, 9600, 115200 etc.



5 : DATABITS

Possible values: 5,6,7,8.

6 : PARITY

O=Oven, E=Even.

7 : STOPBITS

Allowed values: 1 or 2.

8 : FLOWCONTROL

H=Hardware(CTS/RTS) S=Software(XON/XOFF).

9 : INPUT_TIMEOUT (in ms)

You can enter a time-out duration in ms here. This is the time period that is waited after each byte coming from the device (UART) before the data is sent to the other side of the bridge. If 0 is entered, the data is forwarded as quickly as possible. This can be used to collect the data so that it is sent in one piece.

10 : RESET_TIMEOUT(in seconds)

If a device cannot be initialised, the bridge is reset after this time has elapsed and the device is initialised again.

**NET DEVICE Parameter:**

The screenshot shows the XTTelnet application window. At the top, there is a title bar with the text 'XTTelnets'. Below the title bar, there is a menu bar with 'Exit' and 'Connect/Disconnect' buttons. To the right of the menu bar, there is a 'Remote:' label followed by two input fields: '192.168.100.10' and '23'. The main area of the window displays the configuration for 'DEVICE_1' in a text-based format:

```
===== DEVICE_1 =====
1 : TYPE           = NET
2 : DESC           = TCPSERVER 1002
3 : EMULATION      = TCPSERVER
4 : PORT           = 1002
5 : CLIENT_MODE    = SINGLE
6 : CONNECT_TYPE   = AUTO
7 : INPUT_TIMEOUT  =
8 : RESET_TIMEOUT  =
9 : DESTINATIONS   =
```

Below the configuration list, there is a dashed line followed by the text '[Q = QUIT] Please enter your choice:'. At the bottom of the window, there is an 'Input:' label followed by an empty text box. Below the input box, there is a 'State:' label followed by the text 'Verbunden.'.

1 : TYPE

Device type, the following values are permitted: NET, UART, USB, USBUART, TTLIO, SPI, I2C. Other types can be realised on request. Note: If you change a configured type, all current settings are deleted, as each type requires its own configuration parameters.

2 : DESC

Description for this configuration.

3 : EMULATION

Here you can enter the desired operating mode for the respective interface. The following values are permitted: TCPSERVER, UDPSEVER, TCPCLIENT, UDPCLIENT.

4 : PORT

For TCPSERVER and UDPSEVER, the port on which the server is started is entered here.

**5 : CLIENT_MODE(MULTI/SINGLE)**

With TCPCLIENT emulation, you can select the client mode here. MULTI means that the connection is established to all destinations. With SINGLE, the destination reached first always wins.

With UDPCLIENT, only MULTI really makes sense, as the UDP protocol by definition does not provide for a connection to be established in advance. The data is simply sent.

6 : CONNECT TYPE(AUTO/ONDATA)

With TCPCLIENT emulation, you can decide here when the connection is established. With AUTO, the connection is established immediately as soon as the device is ready for operation. With ONDATA, the connection is only established as soon as the first data is received from the interface.

7 : INPUT_TIMEOUT (ms)

You can enter a time-out duration in ms here. This is the time period that is waited after each byte coming from the device (network socket) before the data is sent to the other side of the bridge. If 0 is entered, the data is forwarded as quickly as possible. This can be used to collect the data so that it is sent in one piece.

8 : RESET_TIMEOUT(in Sekunden)

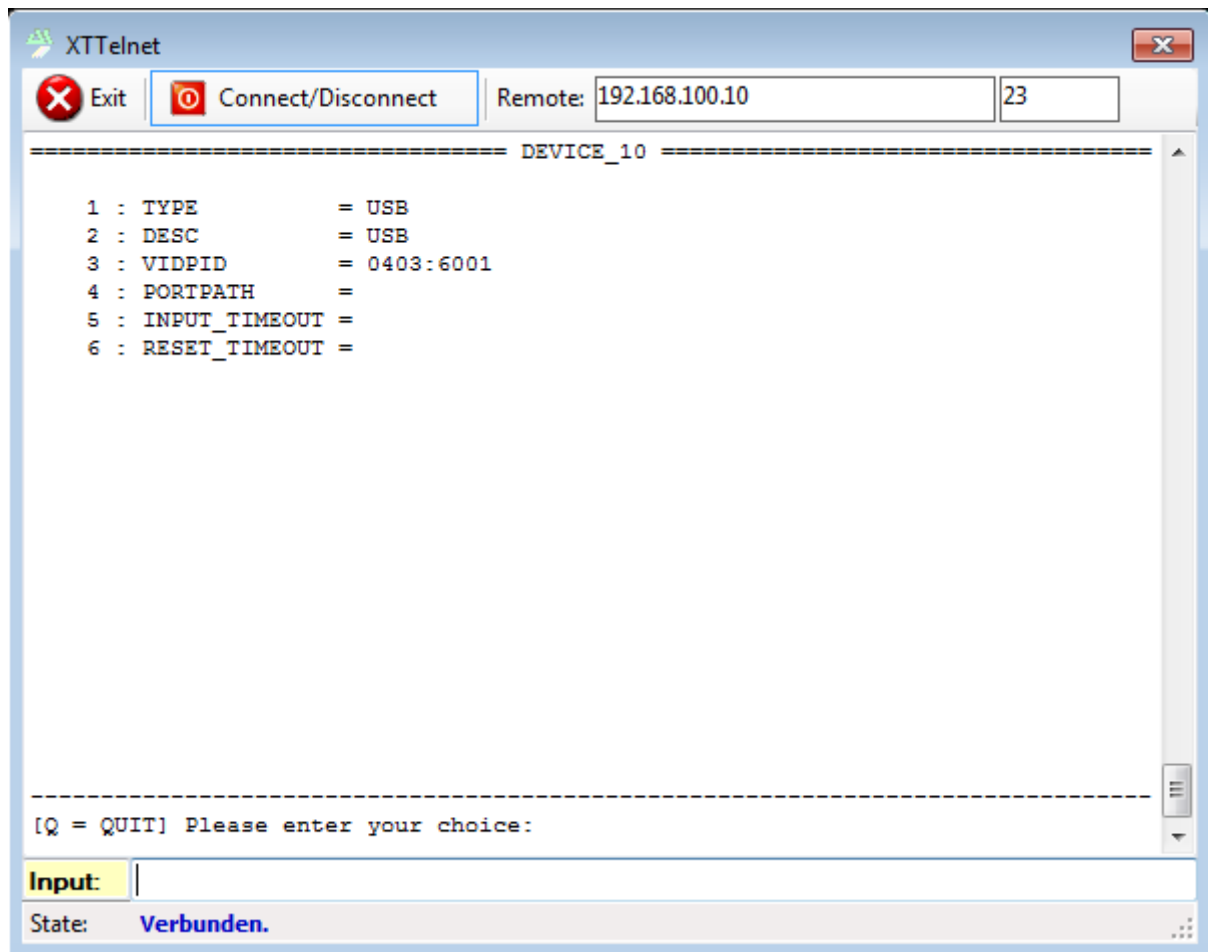
If the device cannot be initialised, the bridge is reset after this time has elapsed and the device is initialised again.



9. DESTINATIONS

The screenshot shows the XTtelnnet application window. At the top, there is a title bar with the text 'XTtelnnet'. Below the title bar, there is a menu bar with 'Exit' and 'Connect/Disconnect' buttons. To the right of the menu bar, there are two input fields: 'Remote: 192.168.14.105' and '23'. The main area of the window displays the text '===== DESTINATIONS =====' followed by 'Page: 1 / 1'. Below this, there is a list of 15 numbered items, each followed by an equals sign: '1 = 192.168.14.240:12345', '2 =', '3 =', '4 =', '5 =', '6 =', '7 =', '8 =', '9 =', '10 =', '11 =', '12 =', '13 =', '14 =', '15 ='. At the bottom of the main area, there is a dashed line followed by the text '[Q = QUIT] Please enter your choice:'. Below this, there is an 'Input:' label and an empty text box. At the very bottom, there is a 'State:' label followed by the text 'Verbunden.'.

Here you can enter the destinations (IP:Port) for the TCPCLIENT or UDPCLIENT emulations. Depending on the CLIENT_MODE, one or more network connections are then established.

**USB DEVICE Parameter:****1 : TYPE**

Device type, the following values are permitted: NET, UART, USB, USBUART, TTLIO, SPI, I2C. Other types can be realised on request. Note: If you change a configured type, all current settings are deleted, as each type requires its own configuration parameters.

2 : DESC

Description for this configuration.

3 : VIDPID

You have the option of recognising a USB device via its VID and PID combination. This is always recommended if you always use the same USB device/model or, alternatively, you can also store the path of a USB port, see point 4: PORTPATH

4 : PORTPATH

This is the alternative configuration variant to point 3: VIDPID. You can use this to store the path of a USB port. This variant is recommended if you want to configure different USB devices for a bridge. Example: /usb2/2-2

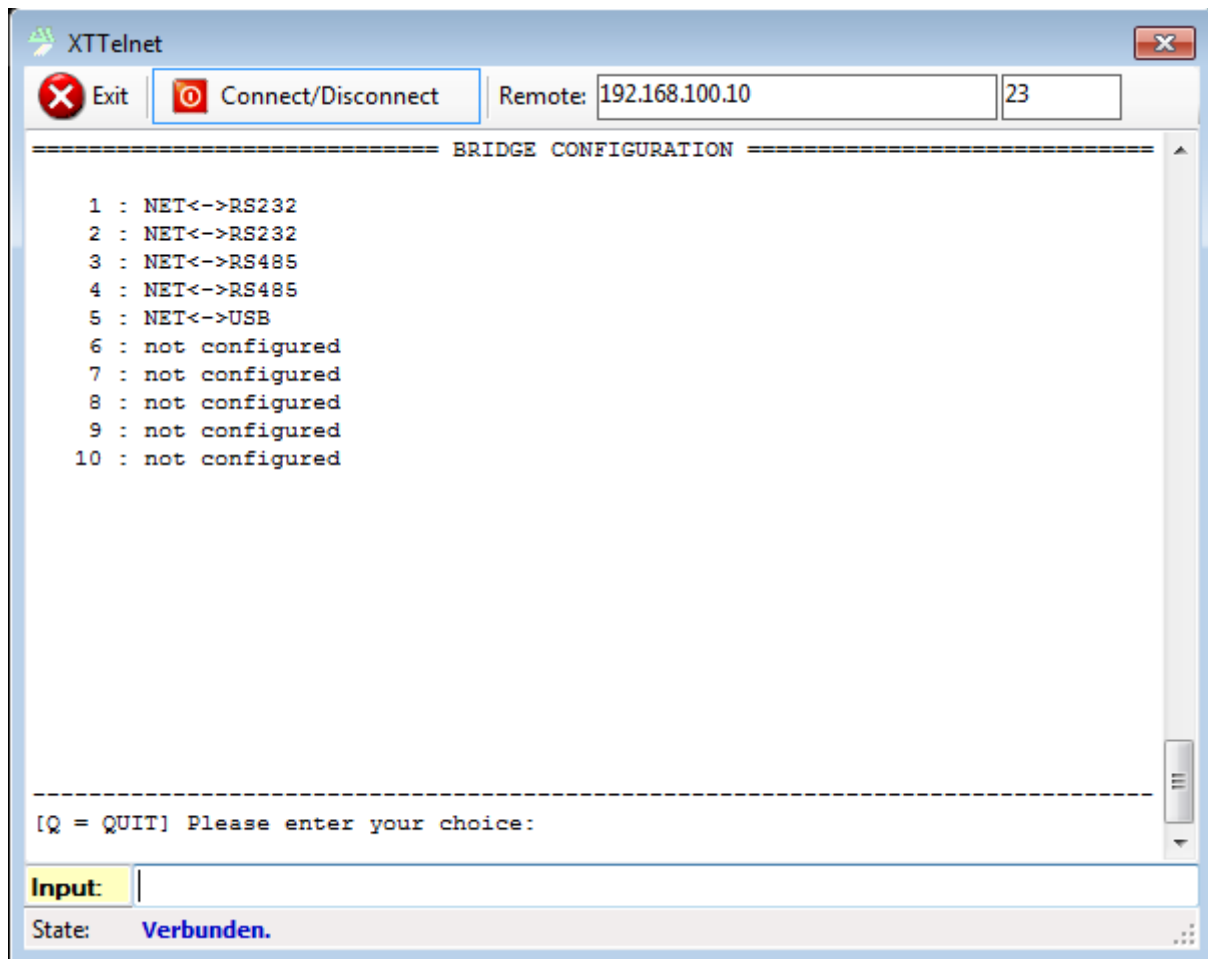
**5 : INPUT_TIMEOUT (ms)**

You can enter a time-out duration in ms here. This is the time period that is waited after each byte coming from the device (USB) before the data is sent to the other side of the bridge. If 0 is entered, the data is forwarded as quickly as possible. This can be used to collect the data so that it is sent in one piece.

6 : RESET_TIMEOUT(in seconds)

If a device cannot be initialised, the bridge is reset after this time has elapsed and the device is initialised again.

This USB device configuration is an alternative to the VIRTUAL USB solution. If you use "Virtual USB", you must deactivate the USB bridge. The same applies to the other direction. If you want to use USB bridge, you must deactivate the Virtual USB configuration for this port.

**BRIDGE CONFIGURATION:**

Here you can see the list of all bridge configurations.

**BRIDGE Parameter:**

The screenshot shows the XTNet application window. At the top, there is a title bar with the text 'XTNet'. Below the title bar, there is a menu bar with 'Exit' and 'Connect/Disconnect' buttons. To the right of the menu bar, there is a 'Remote:' label followed by a text box containing '192.168.100.111' and a port box containing '23'. The main area of the window displays the configuration for 'BRIDGE_1' as follows:

```
===== BRIDGE_1 =====  
  
1 : DESC          = NET<->USB  
2 : DEVICE_A      = DEVICE_1  
3 : DEVICE_B      = DEVICE_2  
4 : PROTOCOL_A    =  
5 : PROTOCOL_B    =  
6 : DEBUG_LEVEL   =  
7 : RESET_PORT    =  
8 : ENABLED       = N  
  
-----  
[Q = QUIT] Please enter your choice:  
  
Input:   
State:  Verbunden.
```

1 : DESC

Description for this configuration.

2 : DEVICE_A

Device ID for the A side of the bridge.

3 : DEVICE_B

Device ID for the B side of the bridge.

4 : PROTOCOL_A

A special protocol/customised solutions for the A-side can be activated here.

5 : PROTOCOL_B

A special protocol/customised solutions for the B-side can be activated here.



6 : DEBUG_LEVEL

- 1 : Basic messages
- 2 : Data
- 4 : Connection
- 8 : Protocol handler

7 : RESET_PORT

Here, you can configure a TCP reset port for this bridge, e.g., 9081. The reset server in the ADMIN MENU must be enabled for this to work.

8 : ENABLED

Y=This configuration is active. N=disabled (saves hardware resources).



Logfile Bridge:

If the ENABLED setting briefly changes to Y and then jumps back to N, the configuration of the devices or the bridge is usually incorrect. You have the following options for identifying the error.

Please log in to the device via SSH (username: **admin**, password: **ak-nord**). You can use any tool for this, such as PuTTY.

Each configured bridge has a log file, for BRIDGE_1, it is
/data/logs/bridge_1.log

To view error messages, you can use the Linux command tail.

``tail -f /data/logs/bridge_1.log`` displays the messages in real time. To stop the display, use the keyboard shortcut <CTRL>+c

If the log file is empty or no new messages are appearing, you can increase the DEBUG_LEVEL in the bridge configuration. You can also set a combination of levels; for all output, please set DEBUG_LEVEL to 15 (1+2+4+8):

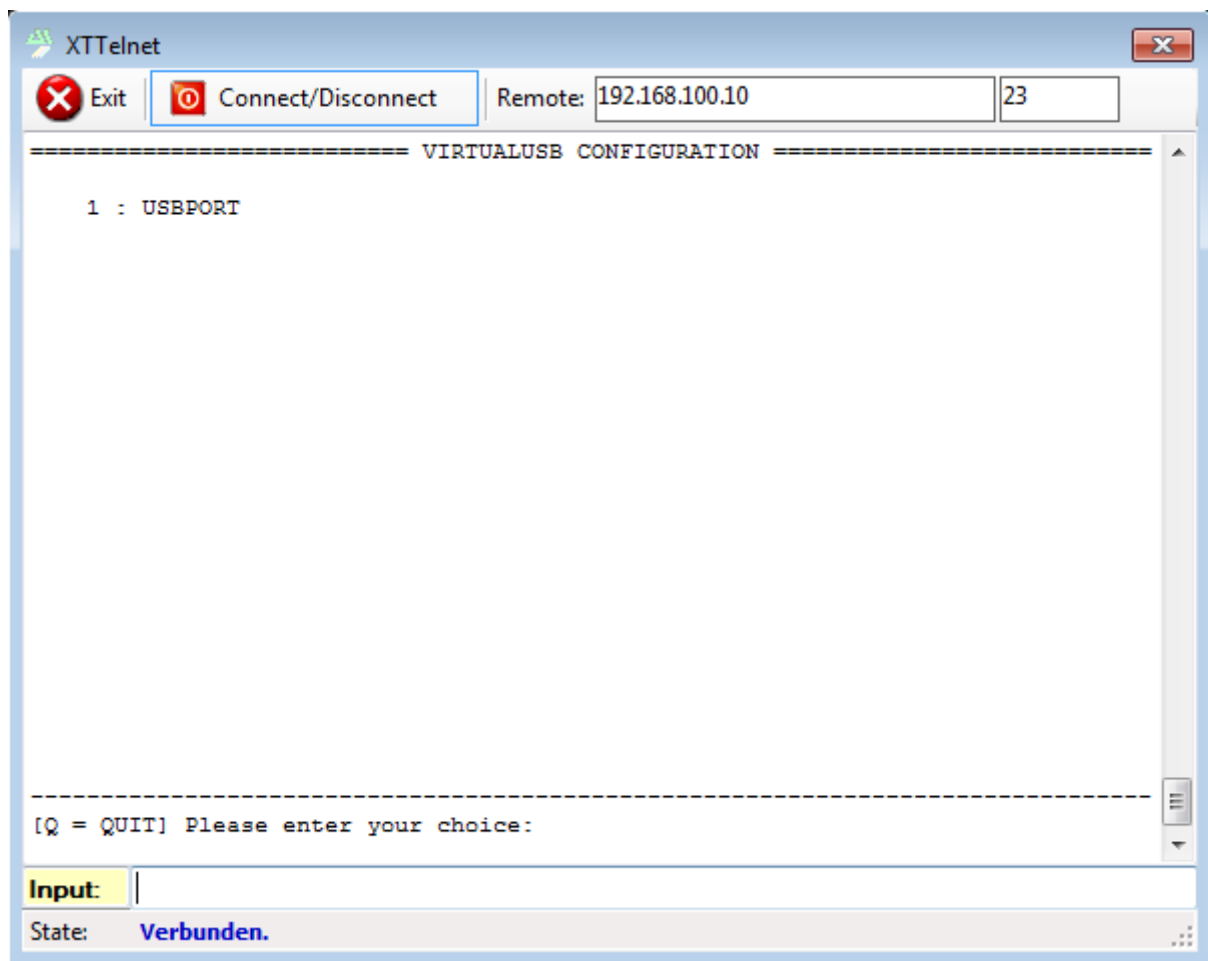
IMPORTANT: Please reset the DEBUG_LEVEL to 0 once you have resolved the issue so that the LOG file does not unnecessarily consume storage space and the Bridge can operate at full speed.

For USB devices, you can also analyze the system log output. Start the log output with this command:

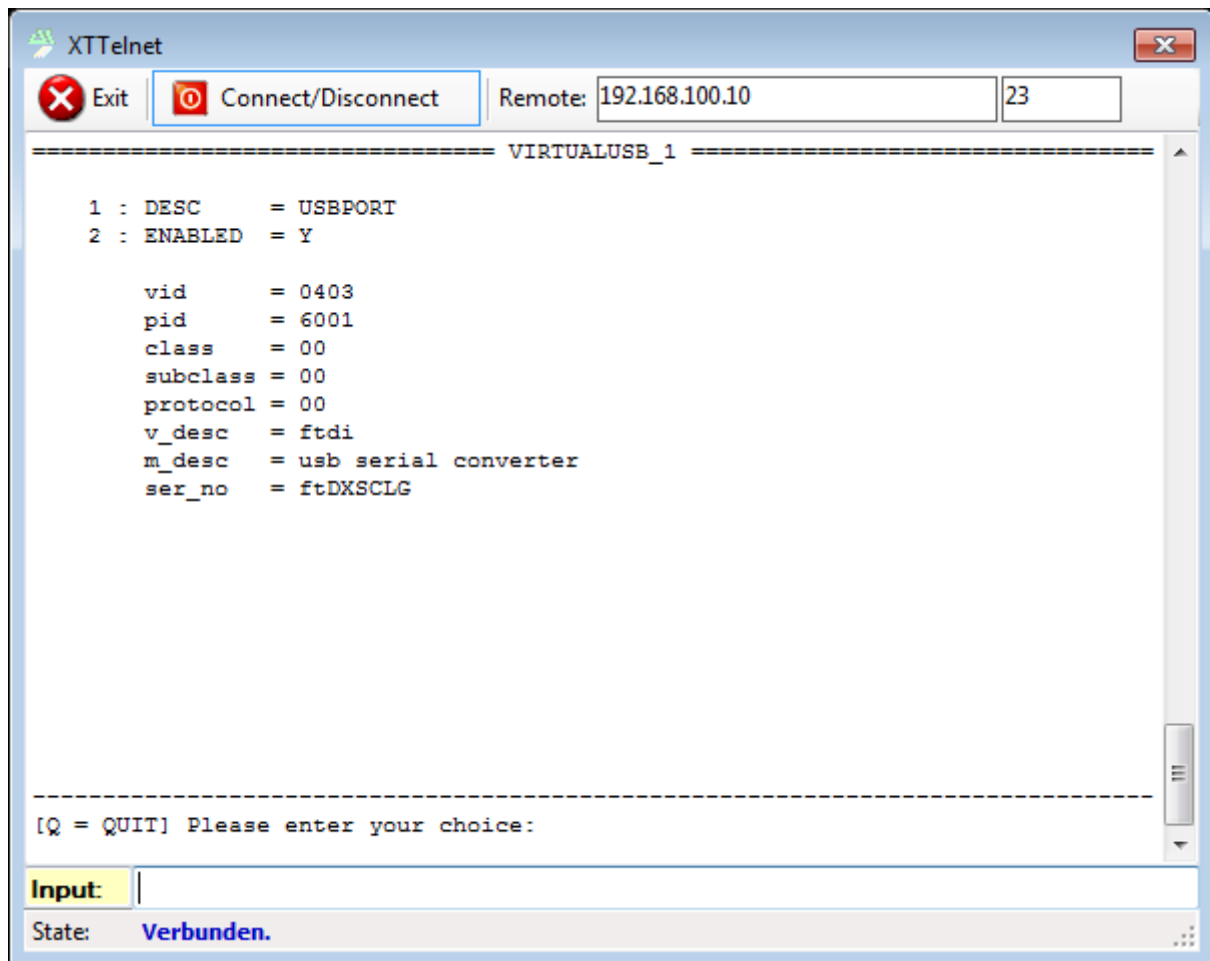
```
tail -f /var/log/messages
```

Then connect your USB device to the AK-Nord product. We classify USB devices

into two different DEVICE types: USB and USBUART. If a device is recognized by Linux as a COM port (e.g., device file /dev/ttyS5 or /dev/ttyUSB1), you must set the device configuration type to USBUART. All other USB devices are configured with the USB type.

**VIRTUALUSB CONFIGURATION:**

Here you can see the list of available USB ports on your device.

**VIRTUALUSB Parameter:****1 : DESC**

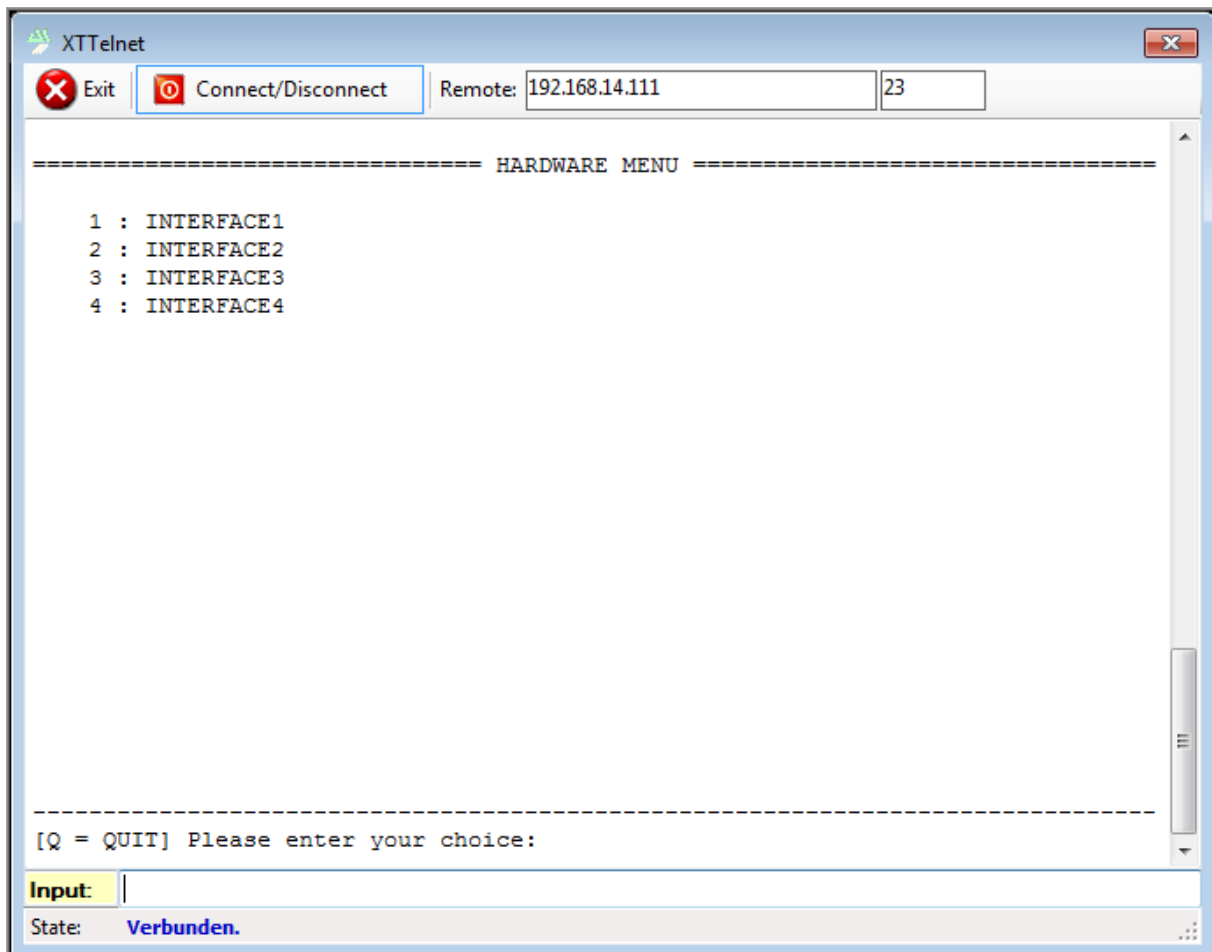
Description for this port. This text is displayed in the graphical user interface of the USB client.

2 : ENABLED

Y=This port is enabled via VIRTUAL-USB and can be controlled by the USB client.

N=This port works without a VIRTUAL-USB solution. This port can therefore be operated via the DEVICE / BRIDGE configuration.

Device information on the currently connected USB device is displayed under the configuration parameters.

**HARDWARE MENU only for ATTO-LXL:**

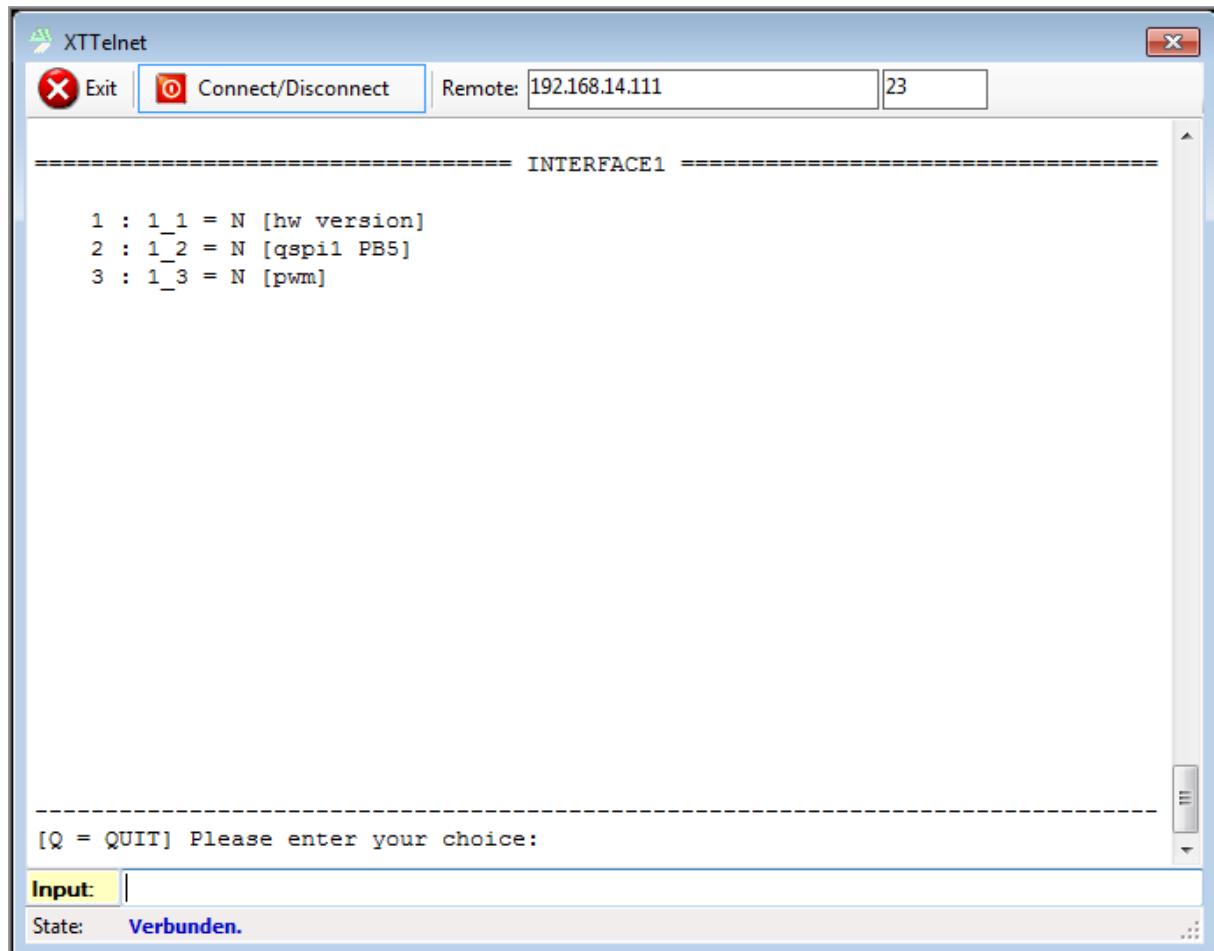
Here you can activate or deactivate the interfaces.
The changes only take effect after a reboot of the ATTO module.

Some interfaces use the same pins and cannot be activated at the same time. Detailed information about the pin assignment of the individual interfaces can be found in the data sheet of ATTO module.



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INTERFACE_1:





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INTERFACE_2:

The screenshot shows a terminal window titled "XTtelnnet". At the top, there are buttons for "Exit" (with a red X icon) and "Connect/Disconnect" (with a red O icon). To the right of these buttons, there are input fields for "Remote:" containing "192.168.14.111" and a port number "23". The main area of the terminal displays a list of components under the heading "INTERFACE2". The components are numbered 1 through 19, each with a label and a description. Below the list, there is a prompt "[Q = QUIT] Please enter your choice:". At the bottom, there is an "Input:" field and a "State:" field showing "Verbunden.".

```
===== INTERFACE2 =====

1 : 2_0   = N [lcd]
2 : 2_1   = N [isc]
3 : 2_3   = N [spi1]
4 : 2_4   = N [pdmic PB11]
5 : 2_5   = N [uart3]
6 : 2_6   = N [qspi1 PB15]
7 : 2_7   = N [i2s1]
8 : 2_8u  = N [fc3 uart]
9 : 2_8s  = N [fc3 spi]
10 : 2_8i  = N [fc3 i2c]
11 : 2_9   = N [pdmic PB26]
12 : 2_10  = N [uart0]
13 : 2_11u = N [fc0 uart]
14 : 2_11s = N [fc0 spi]
15 : 2_11i = N [fc0 i2c]
16 : 2_12  = N [i2c0 PB31]
17 : 2_13  = N [can]
18 : 2_14  = N [i2s0]
19 : 2_15  = N [uart1 DEBUG PC7]

-----
[Q = QUIT] Please enter your choice:

Input:
State:  Verbunden.
```



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INTERFACE_3:

XTTeln

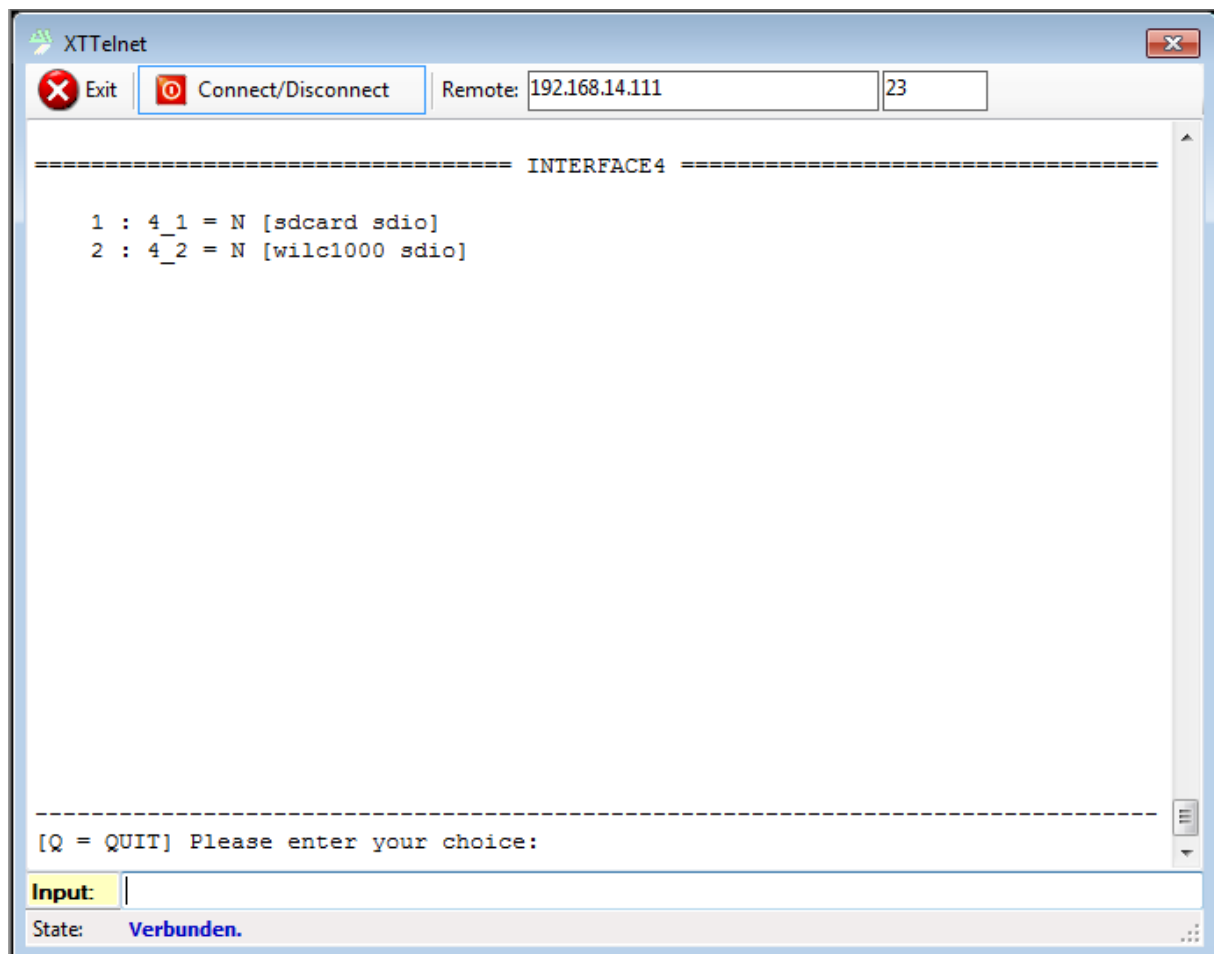
Exit Connect/Disconnect Remote: 192.168.14.111 23

```
===== INTERFACE3 =====  
  
1 : 3_1 = N [spi1]  
2 : 3_2u = N [fc2 uart]  
3 : 3_2s = N [fc2 spi]  
4 : 3_2i = N [fc2 i2c]  
5 : 3_3 = N [uart2]  
6 : 3_3f = N [uart2 8pins]  
7 : 3_4 = N [i2c0 PC18]  
8 : 3_5 = N [adc]  
  
-----  
[Q = QUIT] Please enter your choice:  
  
Input:   
State: Verbunden.
```



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INTERFACE_4:



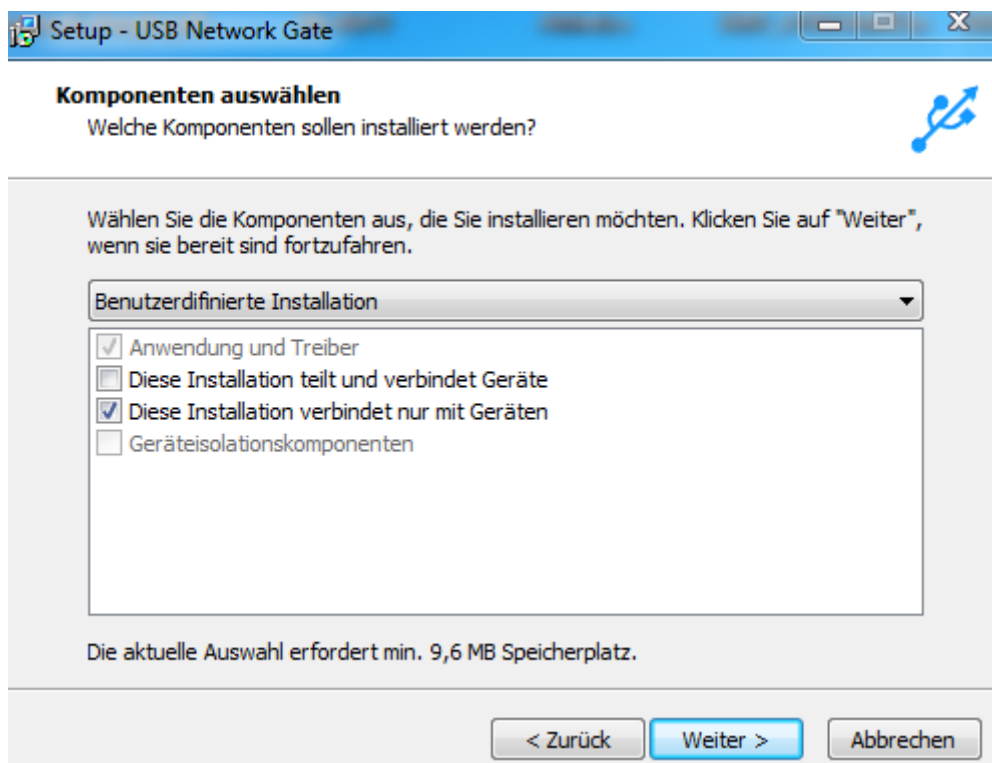


Virtual USB

The Virtual USB service is pre-installed on the device. Please select the emulation "VIRTUAL" at the USB-Config-Menu. To use the virtual USB devices you still need the Virtual-USB-client. You can download it from the this website:

<https://www.eltima.com/de/products/usb-over-ethernet/download/>

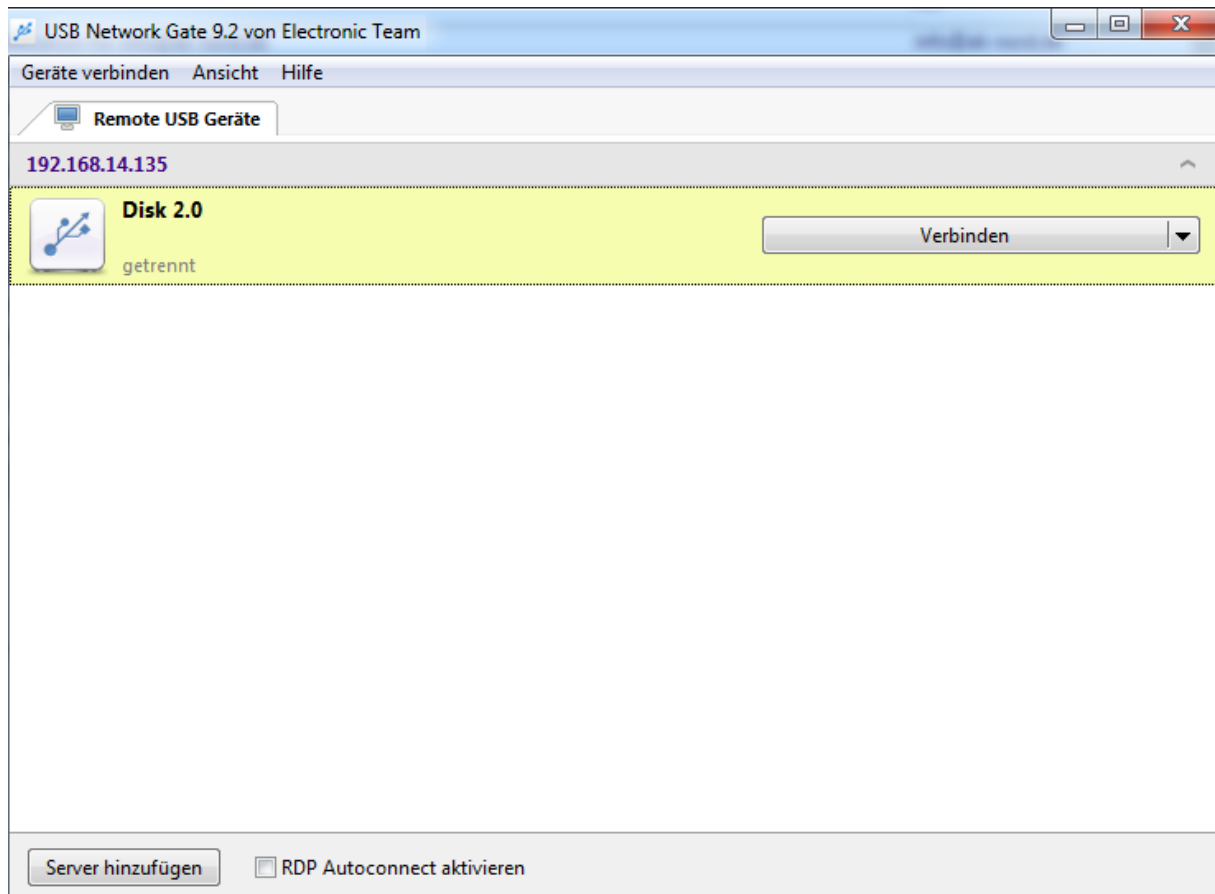
During the installation, please select this option "This installation connects to devices only" in the component selections.



After installing the client, the USB-Server-LXL is displayed in the list of USB hubs and you can connect the connected USB device in the client to the PC.



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Rescue mode

If your device does not boot anymore, you have the possibility to put the device into the reset mode. To do this, press and hold the factory reset button while booting and watch out for the orange LED on the network socket. As soon as the reset mode is started, this LED flashes once for a long time and twice for a short time.

When you are in the Rescue mode, the device gets a fixed IP address in the network: 192.168.100.100.

To install a new firmware you now only have to bring a PC/laptop or similar into the same network. The easiest way is to set the IP address of your PC to e.g. 192.168.100.50 for a short time and then call the following address in the WEB browser:

<https://192.168.100.100:8080>

Here you can then import a new firmware, which you can get from AK-Nord.

To get a firmware please contact:

service@ak-nord.de

Update

There are two ways to update the firmware.

For the first option, you only need a browser and the device must be accessible via the network. Call up the IP address of the device with port 8080 in the browser, e.g:

<http://192.168.100.10:8080>

And install the update via the web interface.

For devices without a network interface, e.g. SER2USB-LXL, you have another option to install the update via a USB stick. To do this, please log in as root via Debug-Console, see chapter "Access via Debug-Console".

Then connect the USB stick where the update file with the name ser2usb.swu is located and then start the script /home/root/usbupdate.sh



Factory Settings

To perform factory reset, you must hold the factory reset button for 5 seconds during operation and then release it.

The factory settings are applied and the device restarts once. Now DHCP is active again and network access is switched on.

Firmware versions:

AK-Nord ships all products with a USB port with firmware that includes a virtual USB solution. Unfortunately, we are not permitted to ship this version with root privileges. With this firmware version, you only have the admin user for SSH access.

If you do not require a Virtual USB solution for your project, we can provide you with an alternative firmware version that does not include the Virtual USB solution but does grant you root access.

In this case, please contact our support team and, ideally, send a screenshot of the Telnet menu login screen. This will allow us to identify your model and current firmware version.

To determine which firmware version is installed on your device, check whether the "VIRTUALUSB CONFIGURATION" menu item is available in the INTERFACE MENU.

AK-DinRail-Server-LXL

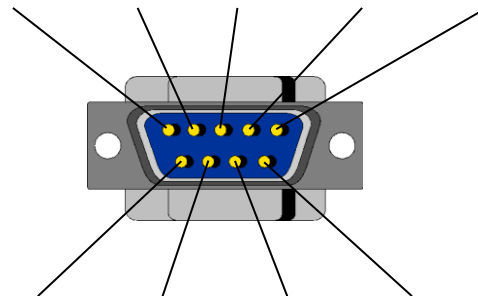


← RS232 interface 2
Port 1003

← RS232 interface 1
Port 1002

RS232 connectors plug

CD(1) RXD(2) TXD(3) DTR(4) GND(5)



DSR(6) RTS(7) CTS(8) RI(9)



← Network

← RS485 connector 2
Port 1005

← RS485 connector 1
Port 1004

← Power supply

RS485 connectors

Ground / GND	
RS485+ / RS485-A	
RS485- / RS485-B	

Power supply

Shield	
+7-60 VDC 2 W	
-7-60 VDC 2 W	



← USB Port

← LED

← DIP-switch RS485
Termination, Pullup, Pulldown

- ① Power LED
- ② Network Link
- ③ Network activity
- ④ Status LED